

PINE RIDGE, SD, USA

WMO: 726517

Lat: 43.021N

Lon: 102.518W

Elev: 3331

StdP: 13.01

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 94039

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-8.7	-2.4	-14.7	2.8	-8.0	-8.8	3.9	-2.1	26.9	30.7	24.6	31.0	2.7	240	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.9	97.7	66.7	94.1	66.2	90.9	65.8	71.3	87.3	69.8	86.0	68.4	84.5	11.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	110.1	77.6	64.6	103.2	76.5	63.3	98.5	75.3	37.2	86.7	35.8	85.8	34.6	84.5	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	23.2	19.8	DB	-22.2	102.8	6.4	3.5	-26.8	105.3	-30.5	107.3	-34.1	109.3	-38.8	111.8	(4)
(5)			WB	-22.3	74.9	6.1	2.6	-26.7	76.8	-30.3	78.3	-33.7	79.8	-38.1	81.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	24.6	25.6	37.4	45.8	55.1	66.6	74.9	72.1	62.5	47.3	35.4	24.6
	DBStd	20.41	11.77	12.49	11.76	9.86	9.18	7.95	6.42	6.87	9.30	9.98	11.29	11.94
	HDD50	3533	786	684	410	188	52	0	0	0	10	167	448	788
	HDD65	7060	1251	1103	854	575	326	74	6	20	157	552	889	1253
	CDD50	2730	0	0	21	64	212	498	771	684	386	84	10	0
	CDD65	781	0	0	0	1	20	121	313	239	83	4	0	0
	CDH74	10963	0	0	22	92	400	1722	4094	3129	1337	156	11	0
	CDH80	5232	0	0	1	12	126	743	2162	1563	582	41	2	0
Wind	WSAvg	8.3	7.2	7.7	8.8	9.9	9.5	9.0	8.7	8.3	8.4	8.0	7.2	7.0
Precipitation	PrecAvg	17.2	0.4	0.5	1.0	2.4	2.8	3.0	1.9	1.7	1.5	1.2	0.7	0.4
	PrecMax	23.2	1.1	1.2	1.9	4.6	7.2	6.0	3.7	3.5	4.1	3.3	2.5	1.3
	PrecMin	12.2	0.1	0.2	0.2	0.6	0.8	1.1	0.2	0.2	0.4	0.2	0.0	0.0
	PrecStd	3.2	0.3	0.3	0.4	1.0	1.7	1.4	0.9	1.0	1.0	1.0	0.5	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	63.1	77.5	81.9	89.8	99.3	103.4	100.1	96.2	85.6	74.5	58.5
		MCWB	42.4	45.4	51.7	54.9	63.0	65.8	67.3	66.4	63.9	57.6	50.8	43.5
	2%	DB	51.5	55.5	70.5	76.6	83.4	92.8	99.0	96.7	91.3	78.7	66.2	51.3
		MCWB	40.1	42.0	49.5	53.9	61.4	65.7	67.4	66.2	62.4	55.2	47.7	39.5
	5%	DB	45.7	48.9	63.8	71.1	78.5	88.2	95.0	93.0	87.5	72.4	60.8	46.1
		MCWB	36.7	38.8	47.2	51.9	59.2	66.2	67.1	65.8	61.6	53.3	44.9	37.2
	10%	DB	41.5	43.9	57.3	64.8	73.2	83.6	91.2	89.6	82.0	66.3	54.2	41.5
		MCWB	34.7	35.6	43.4	49.4	57.1	64.4	67.3	65.9	60.2	50.8	42.1	34.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.6	46.7	54.3	58.0	66.4	72.8	74.2	71.9	69.1	60.0	52.0	44.2
		MCDB	55.9	61.5	72.6	75.6	82.9	86.8	89.7	88.4	85.5	78.2	71.7	57.3
	2%	WB	40.4	42.5	50.3	55.1	63.3	69.7	72.0	70.1	66.1	56.9	48.5	40.2
		MCDB	50.6	54.4	66.9	72.7	78.7	83.8	88.2	87.0	82.5	74.0	65.4	50.0
	5%	WB	37.4	39.3	47.7	52.8	60.7	67.7	70.5	68.8	64.0	54.3	45.5	37.4
		MCDB	45.1	48.1	63.0	69.0	74.9	83.0	86.8	85.5	81.0	69.1	59.4	45.9
	10%	WB	34.9	36.0	44.4	50.4	58.4	65.8	69.1	67.4	62.0	51.8	42.4	34.7
		MCDB	41.0	43.1	57.0	63.9	71.2	81.7	85.3	83.6	77.9	65.0	53.4	41.3
Mean Daily Temperature Range	5% DB	MDBR	24.7	24.8	26.9	26.0	25.8	27.4	28.9	29.7	30.6	28.7	27.7	24.3
		MCDBR	28.8	31.5	37.5	36.9	33.9	32.8	33.6	34.1	36.0	39.0	37.9	29.4
		MCWBR	19.2	19.6	20.0	18.2	15.7	12.9	11.0	11.4	13.5	19.1	21.3	19.6
	5% WB	MCDBR	27.6	29.7	35.3	34.0	29.7	28.9	28.7	29.3	30.7	33.8	36.2	28.7
		MCWBR	18.9	19.1	19.2	17.4	14.7	13.2	11.7	11.3	12.6	17.2	20.6	19.3
Clear-Sky Solar Irradiance	taub	0.254	0.261	0.283	0.314	0.335	0.350	0.371	0.367	0.327	0.290	0.264	0.253	
	taud	2.571	2.554	2.533	2.437	2.453	2.432	2.408	2.400	2.497	2.572	2.559	2.553	
	Ebn at Noon	290	305	308	303	297	292	284	282	287	288	282	279	
	Edh at Noon	21	25	29	34	35	36	36	35	30	24	21	19	
All-Sky Solar Radiation	RadAvg	646	940	1315	1636	1944	2260	2263	1947	1558	1044	722	544	
	RadStd	39	56	101	92	157	149	74	75	87	98	43	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	-2.36	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page